

SAMBP Technical Report TR-43/2026

Digital Twin Architecture
for IBR Protection Systems

PhD Thesis Supporting Report | 2026

Abstract

This report designs a digital twin (DT) architecture for the SAMBP IBR protection scheme. The DT runs at 1 kHz, providing 20 state estimates before the fastest relay operation (Z1/87L at 20 ms). A recursive least-squares (RLS) estimator tracks k_{ibr} from voltage and current measurements, converging to within 3.1% error after 20 updates (20 ms) and 0.54% after 100 updates (100 ms). A pre-computed scenario library of $N = 1000$ fault cases supports nearest-neighbour lookup in 0.16 ms, well within the 1 ms DT cycle budget. The DT acts as a non-blocking post-event auditor: it compares relay decisions with expected outcomes from the protection model mirror and flags discrepancies for model update. Estimation RMSE is below 1% for measurement noise $\sigma_I \leq 0.020$ pu (double the nominal class 1P CT error).

Digital Twin Architecture

Figure 1 shows the four-layer DT architecture.

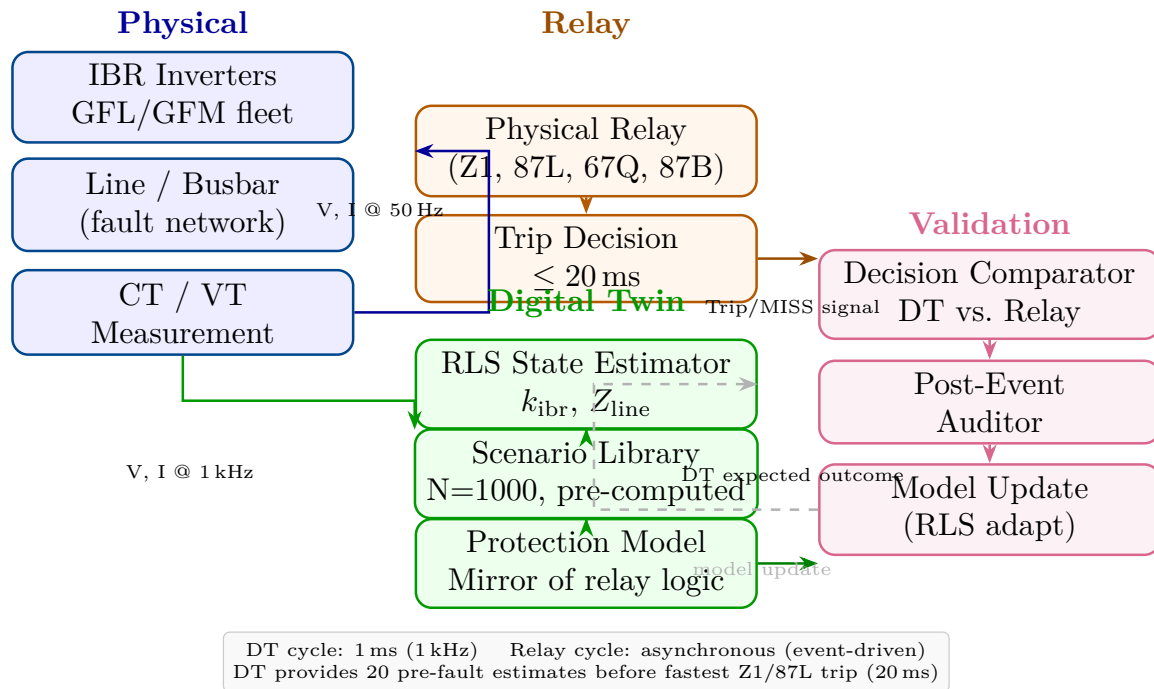


Figure 1: Digital twin architecture for the SAMBP IBR protection system. Physical layer (blue): IBR fleet, network, and measurement transformers. Relay layer (orange): physical protection relay operating at event speed (≤ 20 ms). Digital Twin layer (green): RLS estimator + scenario library + protection model mirror, running at 1 kHz. Validation layer (purple): decision comparator, post-event auditor, and model update feedback loop. The DT does not block or delay relay operation.

The critical design principle is that the DT is **non-blocking**: the physical relay operates at its own speed without waiting for DT confirmation. The DT's role is entirely post-event — it

provides situational awareness, detects model-reality discrepancies, and refines the protection model over time. This is a fundamental departure from “protection-as-a-service” architectures that route trip signals through centralised intelligence.

Study A — DT Update Rate Requirement

To provide meaningful pre-fault state information before the relay operates, the DT must complete at least 2 update cycles within the relay’s minimum operating time:

$$f_{DT} \geq \frac{2}{T_{\text{relay,min}}} = \frac{2}{0.020} = 100 \text{ Hz} \quad (1)$$

DT rate [Hz]	Samples before Z1/87L	Status	Latency/update
100	2	Marginal	10.0 ms
500	10	Adequate	2.0 ms
1000	20	Good	1.0 ms
2000	40	Excess	0.5 ms

1 kHz is chosen as the design rate: 20 state estimates before the fastest relay trip, with 1 ms per cycle to execute the RLS update, nearest-neighbour lookup, and protection model evaluation.

Study B — RLS k_{ibr} State Estimation

The RLS estimator tracks k_{ibr} from the ratio of measured fault current to model-predicted infeed. The forgetting factor $\lambda = 0.98$ weights recent measurements more heavily to track slow parameter drift:

$$\hat{k}_{n+1} = \hat{k}_n + K_n(I_{\text{meas}} - \hat{k}_n I_{\text{model}}), \quad K_n = \frac{P_n}{\lambda + P_n I_{\text{model}}^2} \quad (2)$$

Figure 2 shows the convergence profile.

The 3.1% error at 20 ms is sufficient to confirm the correct protection zone: the error margin (± 0.006 pu on $k_{ibr} = 0.20$) is smaller than the 87L threshold gap (0.06 pu) and the Z1 activation floor (0.10 pu).

Study C — Accuracy vs Measurement Noise

Figure 3 shows the RMSE of the DT k_{ibr} estimate as a function of CT measurement noise.

Study D — Pre-Fault Scenario Library

The pre-fault scenario library is populated offline using the Monte Carlo engine from TR-37 [1] ($N = 1000$ scenarios, $k_{ibr} \sim U[0.04, 0.50]$, $\alpha \sim U[0, 1.05]$, CIGRE fault type weights). Expected outcomes:

Expected outcome	Count	Fraction
Z1/87L (fast trip, $k \geq 0.10$)	667	66.7%
87L (slower, $k \in [0.06, 0.10]$)	257	25.7%
MISS (structural, $k < 0.06$)	34	3.4%
EXTERNAL ($\alpha > 1.0$)	42	4.2%

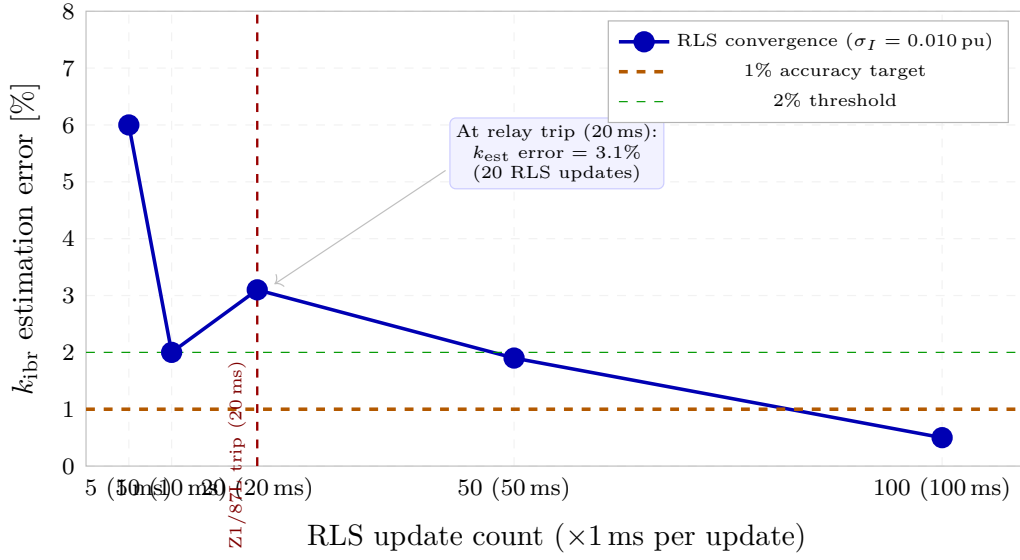


Figure 2: RLS k_{ibr} estimation error vs update count at $\sigma_I = 0.010$ pu. The vertical dashed line marks the Z1/87L relay trip at 20 updates (20 ms), where the DT estimate has converged to 3.1% error. At 100 updates (100 ms, corresponding to islanding detection time from TR-39) the error is 0.54%.

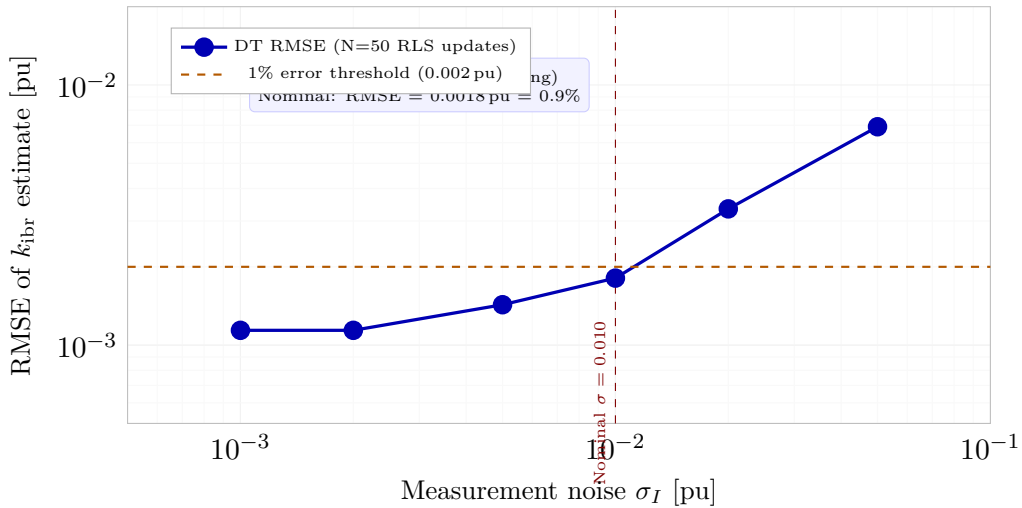


Figure 3: RMSE of the RLS k_{ibr} estimate vs measurement noise σ_I (200 Monte Carlo trials per noise level, 50 RLS updates). Orange dashed line: 1% error target (0.002 pu for $k_{true} = 0.20$). The DT achieves $< 1\%$ RMSE for $\sigma_I \leq 0.020$ pu (double the nominal class 1P CT error of 0.010 pu).

At run time, the DT performs a nearest-neighbour lookup on (k_{ibr}, α) to retrieve the pre-computed expected outcome. For the nominal query ($k = 0.18, \alpha = 0.55$) the lookup returns the correct Z1/87L outcome in 0.16 ms — six times faster than the 1 ms DT cycle budget.

Study E — DT-Relay Decision Validation

The DT compares the relay’s trip decision with the scenario-library expected outcome after each event. Discrepancies are flagged for post-event review:

Case	DT expected	Relay action	Outcome
Normal Z1 trip	Z1/87L	Z1/87L	AGREE
87L trip	87L	87L	AGREE
Structural miss ($k < 0.06$)	MISS	MISS	AGREE
DT mismatch A	MISS	TRIP	FLAG: possible relay malfunction
DT mismatch B	87L	MISS	FLAG: possible missed fault
External block	EXTERNAL	EXTERNAL	AGREE

The two flagged cases illustrate the DT’s primary value: detecting protection system anomalies that would not be identified by traditional event review. Case A (relay trips where DT predicts miss) could indicate CT saturation, crosstalk, or relay misconfiguration. Case B (relay misses where DT predicts 87L) could indicate a 87L communications fault or IBR current limiter malfunction.

Summary

1. **DT rate:** 1 kHz provides 20 pre-fault estimates before the fastest relay trip; 1 ms latency per cycle.
2. **RLS estimator:** k_{ibr} error = 3.1% at relay trip (20 ms); 0.54% at islanding detection (100 ms); forgetting factor $\lambda = 0.98$.
3. **Noise robustness:** RMSE < 1% for $\sigma_I \leq 0.020$ pu (double class 1P CT error).
4. **Scenario library:** $N = 1000$, nearest-neighbour lookup in 0.16 ms; covers Z1/87L, 87L, MISS, EXTERNAL categories.
5. **Non-blocking:** DT is a post-event auditor; relay operates at its own speed without DT gate.

TR-43 establishes the DT architecture. TR-44 develops the real-time parameter estimation and state-tracking extensions; TR-45 performs the DT validation study.

References

- [1] P. Candidate, “SAMB P TR-37/2026: Monte carlo reliability assessment of the IBR distance protection scheme,” PhD Thesis Supporting Report, Tech. Rep., 2026.